

Derivation of a Cam Profile for a Statically Balanced Linear Spring Compensator

Gustavo Torres

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1. Problem Statement

The objective is to design a statically balanced gravity compensator using a payload arm and a linear spring governed by a cam. The system must remain in perfect static equilibrium at any angle θ , requiring zero holding torque.

Known parameters:

Payload mass	m	$= 0.485 \text{ kg}$
Arm length	L	$= 5 \text{ in (0.127 m)}$
Spring constant	k	$= 974.1 \text{ N/m}$
Gravity	g	$= 9.81 \text{ m/s}^2$

Unknown: The cam radius profile $R(\theta)$ required to maintain equilibrium.

2. Potential Energy Formulation

For a statically balanced system the total potential energy V must be constant with respect to θ :

$$\frac{\partial V}{\partial \theta} = 0 \quad (1)$$

The total potential energy combines gravitational (V_g) and elastic (V_s) contributions,

$$V = V_g + V_s, \quad (2)$$

where $\theta = 0$ denotes Top-Dead-Center (arm pointing vertically upward). Explicitly:

$$V_g = mgL \cos \theta, \quad (3)$$

$$V_s = \frac{1}{2}kx^2, \quad x(\theta) = \int_0^\theta R(\alpha) d\alpha. \quad (4)$$

3. Torque Balance and the Cam Profile

Differentiating V and setting the result to zero gives the torque balance condition:

$$\frac{\partial V}{\partial \theta} = -mgL \sin \theta + kx \frac{dx}{d\theta} = 0 \implies mgL \sin \theta = k \cdot x(\theta) \cdot R(\theta). \quad (5)$$

To satisfy (5) identically over a full 360° sweep, we propose:

$$R(\theta) = R_{\max} \cos\left(\frac{\theta}{2}\right). \quad (6)$$

Integrating to obtain the spring displacement:

$$x(\theta) = \int_0^\theta R_{\max} \cos\left(\frac{\alpha}{2}\right) d\alpha = 2R_{\max} \sin\left(\frac{\theta}{2}\right). \quad (7)$$

Substituting back into the spring torque and applying the double-angle identity $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$:

$$\tau_s = k \left(2R_{\max} \sin \frac{\theta}{2} \right) \left(R_{\max} \cos \frac{\theta}{2} \right) = k R_{\max}^2 \sin \theta. \quad (8)$$

Equating $\tau_g = \tau_s$ and cancelling $\sin \theta$ yields the closed-form result:

$$\boxed{R_{\max} = \sqrt{\frac{mgL}{k}}} \quad (9)$$

4. Prototype Calculation

Substituting the prototype values:

$$R_{\max} = \sqrt{\frac{(0.485)(9.81)(0.127)}{974.1}} = \sqrt{\frac{0.6042}{974.1}} = \sqrt{6.203 \times 10^{-4}} \approx 0.0249 \text{ m}. \quad (10)$$

Converting to inches for the CAD model:

$$R_{\max} = 0.0249 \text{ m} \times \frac{1 \text{ in}}{0.0254 \text{ m}} \approx 0.980 \text{ in}. \quad (11)$$

Conclusion. To balance the 0.485 kg mass on a 5-inch arm, the cam profile is:

$$\boxed{R(\theta) = 0.980 \cos\left(\frac{\theta}{2}\right) \quad [\text{inches}]} \quad (12)$$

5. Cam Profile Coordinates (5-inch Arm)

The cam edge is modelled in CAD by fitting a spline through the Cartesian coordinates below. With the centre axle at the origin, points run from TDC (0) to BDC (180) using $X = r \sin \theta$, $Y = r \cos \theta$, where $r = 0.980 \cos(\theta/2)$. Mirror about the Y -axis for the full profile.

Angle θ (°)	Radius r (in)	X (in)	Y (in)
0	0.980	0.000	0.980
10	0.976	0.169	0.961
20	0.965	0.330	0.907
30	0.947	0.473	0.820
40	0.921	0.592	0.705
50	0.888	0.680	0.571
60	0.849	0.735	0.424
70	0.803	0.754	0.275
80	0.751	0.739	0.130
90	0.693	0.693	0.000
100	0.630	0.620	−0.109
110	0.562	0.528	−0.192
120	0.490	0.424	−0.245
130	0.414	0.317	−0.266
140	0.335	0.215	−0.257
150	0.254	0.127	−0.220
160	0.170	0.058	−0.160
170	0.085	0.015	−0.084
180	0.000	0.000	0.000